Gunjan Kumar 24/10/SC/036

3

SET-A

Discrete Mathematics (Minor Exam)

Max Marks: 20 **Time: 90 minutes**

Attempt any two from questions 1, and 2-5 are compulsory

1. (i) Find the PCNF for the given well-formed formula: 2

$$(\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q)$$

(ii) Let $S = \{x \in \mathbb{R} | 0 \leq x \leq 1\}$ and $T = \{x \in \mathbb{R} | 0.5 \leq x \leq 5\}$. Find $S - T$ and $S \oplus T$. 2

(iii) Show that the "greater than or equal "relation is a partial ordering on a set of positive integers. 2

(iv) If f, g, and h are functions from $\mathbb{R}$ to $\mathbb{R}$ defined by $f(x) = x^2, g(x) = x + 5$, and $h(x) = \sqrt{x^2}$, then verify that $fo(goh)=(fog)oh$. 2

2. What is the maximum height of any AVL tree with 10 nodes? Assume that the height of a tree with a single node is 0. Also construct an AVL tree for the given set of numbers: 8,7,1,2,6,5,3,4. 4

3. Differentiate the least upper bound and greatest lower bound. Analyze the time complexity of the given code:

```
A(n)
{
int i, j, k, n;
For(i=1; i<=n; i++)
{  For(j=1; j<=i^2; j++)
  {   For(k=1; k<=n/2; k++)
   {     Printf("SC&SS");
   }}}}
```

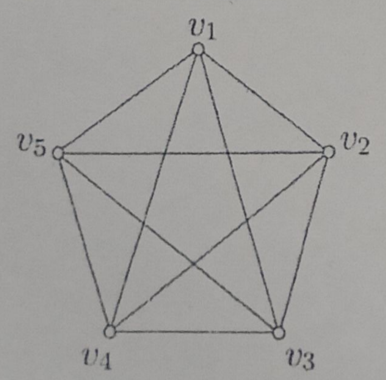
$f(1) = 1^4$

$g(1) = 1 + 5 = 6$

$h(1) = \sqrt{1^2} = 1$

4. Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset." 4

5. Consider a graph G (V, E), Check whether a given graph is bipartite. If it is a bipartite graph then apply the breadth-first search (BFS) using queue data structure otherwise take the counter-example and prove that the bipartite graph is a 2-colorable problem and also show the elements in each layer L_0, $L_1, \ldots, L_k$ produced by the BFS starting from the node v_1, and prove that no edge of G joins two nodes of the same layer. 4



SET-B

Discrete Mathematics (Minor Exam)

Max Marks: 20 **Time: 90 minutes**

Attempt any two from questions 1, and 2-5 are compulsory

1 (i) Find the PCNF for the given well-formed formula: 2

$$(\neg p \wedge \neg q) \rightarrow (p \leftrightarrow \neg q)$$

(ii) Let $S = \{x \in \mathbb{R} | 0 \leq x \leq 0.6\}$ *and* $T = \{x \in \mathbb{R} | 0.3 \leq x \leq 1\}$. *Find* $S - T$ *and* $S \oplus T$.

(iii) Show that the "greater than "relation is a partial ordering on a set of positive integers 2

(iv) If *f, g, and h* are functions from $\mathbb{R}$ to $\mathbb{R}$ defined by $f(x) = x + 3,\ g(x) = x^2$, *and* $h(x) = \sqrt{x^2}$, then verify that *fo(goh)=(fog)oh*. 2

2. What is the maximum height of any AVL tree with 12 nodes? Assume that the height of a tree with a single node is 1. 4
Construct an AVL tree for the given set of numbers: 7,8,2,1,5,6,4,3.

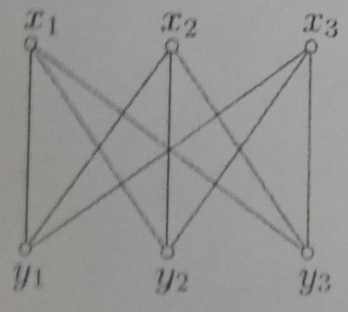
3. Differentiate the least upper bound and greatest lower bound. 4
Analyze the time complexity of the given code:

```
A(n)
{
int i, j, k, n;
For(i=1; i<=n; i++)
{For(j=1; j<=i^2; j++)
  {For(k=1; k<=n; k++)
   {
    Printf("SC&SS");
   }}}}
```

4. 4

Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed."

5. Consider a graph G (V, E), Check whether a given graph is bipartite. If it is a bipartite graph then apply the breadth-first search (BFS) using queue data structure and also show the elements in each layer $L_0,\ L_1, \ldots, L_k$ produced by the BFS starting from the node x_1, and prove that no edge of G joins two nodes of the same layer. 4



CS-404: Discrete Mathematics

Minor-I Test

Date: 25/09/2025 Max Marks: 15

Time: 10:30 – 11:30am

1. List any two utilities of the cardinality of a set for implementation of Set in a computer programme.

2. Given a set say A – {a, e, i, o, u, x, y, z} and r = 3. What is the meaning of nPr?

3. For set complementation as and operation, what is the essential requirement for its implementation in a computer programme?

4. Give the formula and the number of characteristic vectors of a set of all digits if first bit is fixed for digit 3 and last bit for digit 6.

5. If 11 integers are randomly selected from the set {1, 2, ..., 20} then prove that there will be at least a pair a, b such that $a - b = 2$.

6. Give the difference between– Set, Sequence and List and one commonality if any.

7. Give the Predicates required to rewrite the following property as a well-formed formula.

8. Write the instruction/expression which is iterated in a method to calculate GCD of two integers a and b. Why is the method called Euclidean Algorithm?

9. Write the statement of strong form of Mathematical Induction.

10. Consider the following statements – For all books if book is cheap then the book is valuable and for all books if book is costly then the book is valuable. It can be inferred that all books are valuable.
- Represent these statements as wff.
- Which rule of inference can be used to prove the above?

CS-404: Discrete Mathematics

<u>Minor-II Test</u>

Date: 29/11/2025 Max Marks: 15

Time: 10:30 – 11:30am

Answer any THREE of the following. Each question carries 5 marks

Q1. Let f: A → B. Give the conditions when **f** is said to be ***defined everywhere*** and **f** is said to be ***onto.*** Given a set X define and describe identity function on X. *Consider an example of a software* ***E2M*** *similar to Google translate to translate text from English to Marathi and vice-versa. Give any four Discrete Mathematics concepts that are used in this example.*

Q2. Consider a graph G = (V, E). What ae V and E? Give conditions when G is a **graph**, a **regular graph,** and a **complete** graph. *Draw the floor-sketch of a house with two rooms, one kitchen, one toilet-cum-bathroom and one storeroom including its doors. Ignore the windows in the sketch. Represent the floor-sketch as a graph.* Determine if your floor-sketch includes an Euler's Path or an Euler's circuit or both.

Q3. Consider the given set of transition functions defined as δ: Q ⟶Q, {q0, 0= q0, δq0, 1= q1, δq1, 0= q1, δq1, 1= q1}. Give the 5-tuple describing a finite state machine M. Let L be the set of strings accepted by M. Is (L,) a Semigroup or a Monoid where ∘ is string concatenation? Justify your answer.

Q4. Draw the undirected graph $G = (V_1, E_1)$. Let V_1 = {a, b, c, d, e, f, g, h, i} and E_1 = {(a, b), (a, d), (b, c), (c, i), (c, f), (i, f), (f, e), (d, e), (a, g), (g, h), (g, d)}. Give the Hamiltonian path and Hamiltonian circuit if these exist. Given a Hamiltonian graph with k vertices. What is the length of the Hamiltonian path?